1.3 Data storage and compression continued

Candidates should be able to:	Notes and guidance
4 Understand how files are compressed using lossy and lossless compression methods	• Lossy compression reduces the file size by permanently removing data, e.g. reducing resolution or colour depth, reducing sample rate or resolution • Lossless compression reduces the file size without permanent loss of data, e.g. run length encoding (RLE)

2 Data transmission

2.1 Types and methods of data transmission

Candidates should be able to:	Notes and guidance
1 (a) Understand that data is broken down into packets to be transmitted	
(b) Describe the structure of a packet	• A packet of data contains a: – packet header – payload – trailer • The packet header includes the: – destination address – packet number – originator's address
(c) Describe the process of packet switching	• Data is broken down into packets • Each packet could take a different route • A router controls the route a packet takes • Packets may arrive out of order • Once the last packet has arrived, packets are reordered
2 (a) Describe how data is transmitted from one device to another using different methods of data transmission	• Including: – serial – parallel – simplex – half-duplex – full-duplex
(b) Explain the suitability of each method of data transmission, for a given scenario	• Including the advantages and disadvantages of each method
3 Understand the universal serial bus (USB) interface and explain how it is used to transmit data	• Including the benefits and drawbacks of the interface

2.2 Methods of error detection

Candidates should be able to:	Notes and guidance
1 Understand the need to check for errors after data transmission and how these errors can occur	• Errors can occur during data transmission due to interference, e.g. data loss, data gain and data change
2 Describe the processes involved in each of the following error detection methods for detecting errors in data after transmission: parity check (odd and even), checksum and echo check	• Including parity byte and parity block check
3 Describe how a check digit is used to detect errors in data entry and identify examples of when a check digit is used, including international standard book numbers (ISBN) and bar codes	
4 Describe how an automatic repeat query (ARQ) can be used to establish that data is received without error	• Including the use of: – positive/negative acknowledgements – timeout

2.3 Encryption

Candidates should be able to:	Notes and guidance
1 Understand the need for and purpose of encryption when transmitting data	
2 Understand how data is encrypted using symmetric and asymmetric encryption	• Asymmetric encryption includes the use of public and private keys